SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

Database Management Systems: Relational Models & SQL

Department: CSE	**Course / Subject:** Database Management Systems (CS402PC)
Semester: Semester 4	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: Entity-Relationship Diagrams, Relational Algebra, SQL queries, Normalization (1NF through BCNF), Transaction Processing (ACID), and Concurrency Control.

Unit 1: Database Architecture & ER Model

3-tier database architecture, Data independence, Strong & weak entity sets, Cardinality constraints, and Extended ER features.

Unit 2: Relational Algebra & Calculus

Fundamental operators (Select, Project, Cartesian Product, Union, Set Difference), Rename, Joins (Theta, Equi, Natural, Outer), and Tuple Relational Calculus.

Unit 3: SQL & Database Normalization

DDL, DML, DCL, Aggregate functions, Nested subqueries, Functional dependencies, Armstrong axioms, 1NF, 2NF, 3NF, and Boyce-Codd Normal Form (BCNF).

Unit 4: Transaction & Concurrency Control

ACID properties, Serializability (Conflict & View), Two-Phase Locking (2PL), Deadlock prevention and detection, and Timestamp-based protocols.

Unit 5: Storage and Indexing

RAID levels, B-Trees and B+ Trees indexing structure, Static and Dynamic hashing, and Query optimization execution plans.